

PAPER_1304_NEUTRINO_MASS_ABSOLUTE

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PAPER_1304 — Absolute Neutrino Mass Scale

REFINED June 2026: $\Sigma m_{\nu_NH} = \Lambda \times A_5 \times \Phi_res / D_BSFG = 0.00729735 \times 60 \times 0.84 / 6 = 0.0613 \text{ eV}$
vs obs 0.058 (5.7%)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: C — Particle Physics Open (CLOSED)

Date: June 11, 2026

Calculator surface: `calculate_paradox({"paradox": "neutrino_mass_absolute"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1304})`

Master Expression

``

$$\Sigma m_{\nu} = \Lambda \times \Phi_{res} \times (D_{phys} + 1) \times K_{MEX} = 0.0639 \text{ eV}$$

``

Match: 10% from NH min 0.058 eV

Interpretation

$m_{\nu_lightest} = \Lambda \times \Phi_{res} = 6.13 \text{ meV}$. Sum lands in NH-IH band.

UQFF Locked Primitives

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$
- $\Lambda_{QCD} = 0.217 \text{ GeV}$, $v_{Higgs} = 246 \text{ GeV}$ (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

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Subject matter: UQFF derivation of Absolute Neutrino Mass Scale.